

FAULT CODE 638 - Exhaust Gas Temperature Deviation Low for Cylinder 16 - Data Valid But Below Normal Operating Range - Least Severe Level

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the exhaust sensor circuit cylinder 16 accuracy.	
	STEP 1A. Verify the exhaust gas temperature sensor circuit cylinder 16 accuracy.	Temperature measurement from the infrared thermometer within 15 percent of the exhaust gas temperature sensor circuit cylinder 16 reading with INSITE™ electronic service tool?
STEP 2.	Clear the fault code.	
	STEP 2A. Disable the fault code.	Fault Code 638 inactive?
	STEP 2B. Clear any inactive fault codes.	All fault codes cleared?

STEP 1. Check the exhaust gas temperature sensor circuit cylinder 16 accuracy.

STEP 1A. Verify the exhaust gas temperature sensor circuit cylinder 16 accuracy.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
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Verify the exhaust gas temperature sensor circuit cylinder 16 accuracy with an infrared thermometer. • Start the engine. • Connect INSITE™ electronic service tool. • Use an infrared thermometer to measure and record the surface temperature at the exhaust port side of the cylinder head. • Compare the infrared thermometer temperature measurement with the exhaust gas temperature sensor circuit cylinder 16 reading on INSITE™ electronic service tool monitor screen.	Temperature measurement from the infrared thermometer within 15 percent of the exhaust gas temperature sensor circuit cylinder 16 reading with INSITE™ electronic service tool? YES Repair : Possible cylinder or injector damage must be investigated.	Engine Performance Troubleshooting Tree
	Temperature measurement from the infrared thermometer within 15 percent of the exhaust gas temperature sensor circuit cylinder 16 reading with INSITE™ electronic service tool? NO Repair : A damaged exhaust gas temperature sensor has been detected. Replace the exhaust gas temperature sensor. • Refer to Procedure 019-013 in Section 19. (/qs3/pubsys2/xml/en/procedures/81/81-019-013.html)	2A

STEP 2. Clear the fault codes.

STEP 2A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Disable the fault code. • Use INSITE™ electronic service tool to verify the fault code is inactive.	Fault Code 638 inactive? YES	2B
	Fault Code 638 inactive? NO Repair : Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	1A

STEP 2B. Clear the inactive fault codes.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Clear the inactive fault codes. • Use INSITE™ electronic service tool to clear any inactive fault codes.	All fault codes cleared? YES	Repair complete
	All fault codes cleared? NO	Appropriate troubleshooting steps

